

UE23CS320B - Capstone Project Phase - 2

Project Progress Review #2

Project Title : Integrating Polygenic Risk Score and Digital Twin for Coronary Artery Disease Prediction

Project ID : PW26_SVM_06

Project Guide : Dr. Sapna VM

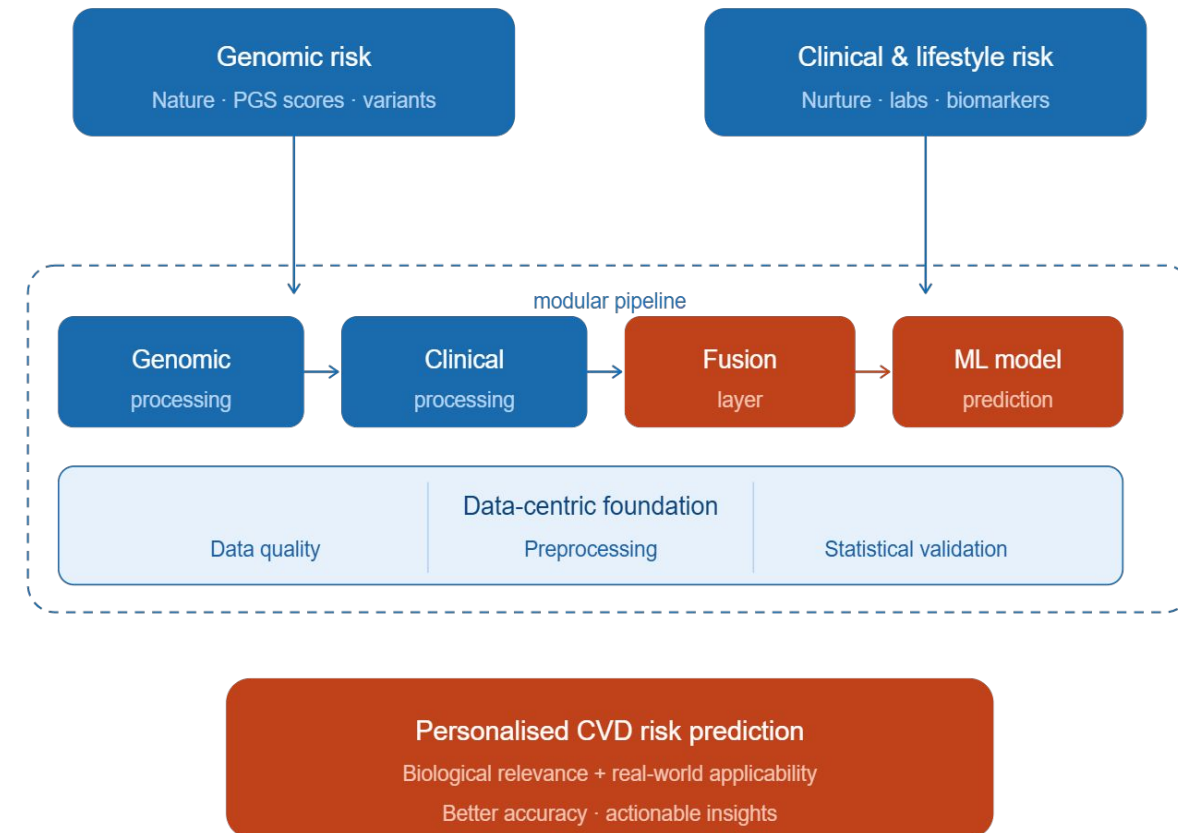
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Problem Statement

- **Problem:** Coronary Artery Disease is the leading cause of death in India. South Asians develop CAD a decade earlier than Western counterparts and face four times the risk — yet every major clinical risk tool in use today was designed and validated on European populations. The result is systematic underdiagnosis at the point when intervention would matter most.
- **The Gap:** Current prediction models treat all patients the same. They use static questionnaires, ignore inherited genetic predisposition, and produce a single risk number with no explanation and no ability to simulate what happens if the patient changes their behaviour. No existing model accounts for the specific genetic architecture of the Indian population.
- **Our Approach** We integrate two data streams, a Polygenic Risk Score calibrated to Indian allele frequencies via Monte Carlo simulation, and a lifestyle dataset of 70k patient records along with 1190 instances of clinical data into a single hybrid Digital Twin framework. The system predicts CAD risk, explains which factors drove the prediction, and enables what-if simulation so a clinician can ask "what changes if this patient stops smoking or controls their BP?"

What is the design approach followed? And Why?

- Hybrid Nature–Nurture Modeling** Genetic data captures inherited risk; lifestyle data captures modifiable factors. The fusion layer combines both for a complete, personalized risk profile.
- Multi-Stage Modular Architecture** Five independent stages: Genomic Processing → Lifestyle Modeling → Fusion → Clinical Enhancement (optional) → ML Prediction.
- Conditional Clinical Integration** Clinical data is optional it is incorporated only when available. The system remains fully functional without it, making it practical for real-world settings where lab tests may not be accessible.
- Digital Twin Risk Fusion** Genetic and lifestyle risk scores are combined to construct a personalized digital twin.
- Data-Centric Validation Pipeline** All inputs — SNP variants, lifestyle variables, clinical biomarkers undergo preprocessing, quality control, and statistical validation before reaching the model.



Benefits of this approach & are there any drawbacks?

Benefits

Higher prediction accuracy
Genomic + clinical fusion

Population-specific modelling
Indian genomic baseline

Modular architecture
Easy to extend or upgrade

Explainable predictions
Using SHAP values

Data-centric approach
Improved reliability & validation

Drawbacks

High-quality genomic data required
Availability may limit deployment

Computationally intensive
Monte Carlo simulation overhead

MICE imputation may introduce bias
Assumes missing at random (MAR)

Integration complexity
Genomic + clinical data alignment

Design Constraints, Assumptions & Dependencies

Discuss the design constraints and assumptions that you have made to select the design approach.

Design Constraints

- Limited availability of paired genomic + clinical data
- Need for population-specific modeling (Indian cohort)
- Presence of missing clinical values
- High computational cost (simulation + ML training)

Design Assumptions

- GenomeIndia frequencies represent Indian population baseline
- SNP effect sizes (PGS) are transferable across populations
- Missing data follows Missing At Random (MAR) (*for MICE*)
- Z-score normalization preserves feature comparability

Design Constraints, Assumptions & Dependencies

Discuss any dependencies that your design approach has and their impact on the project.

Risk level	Dependencies	Impact if it fails
High risk	Data GenomeIndia dataset · PGS / SNP weights Methodological Allele alignment · MICE imputation (MAR assumption)	Invalid or incorrect predictions Wrong population baseline · risk score inversion · biased data
Medium risk	Methodological Monte Carlo simulation · PGS transferability Data Clinical dataset quality (missing values, noise)	Reduced accuracy or generalisation Poor risk distribution (μ, σ) · cross-population limits

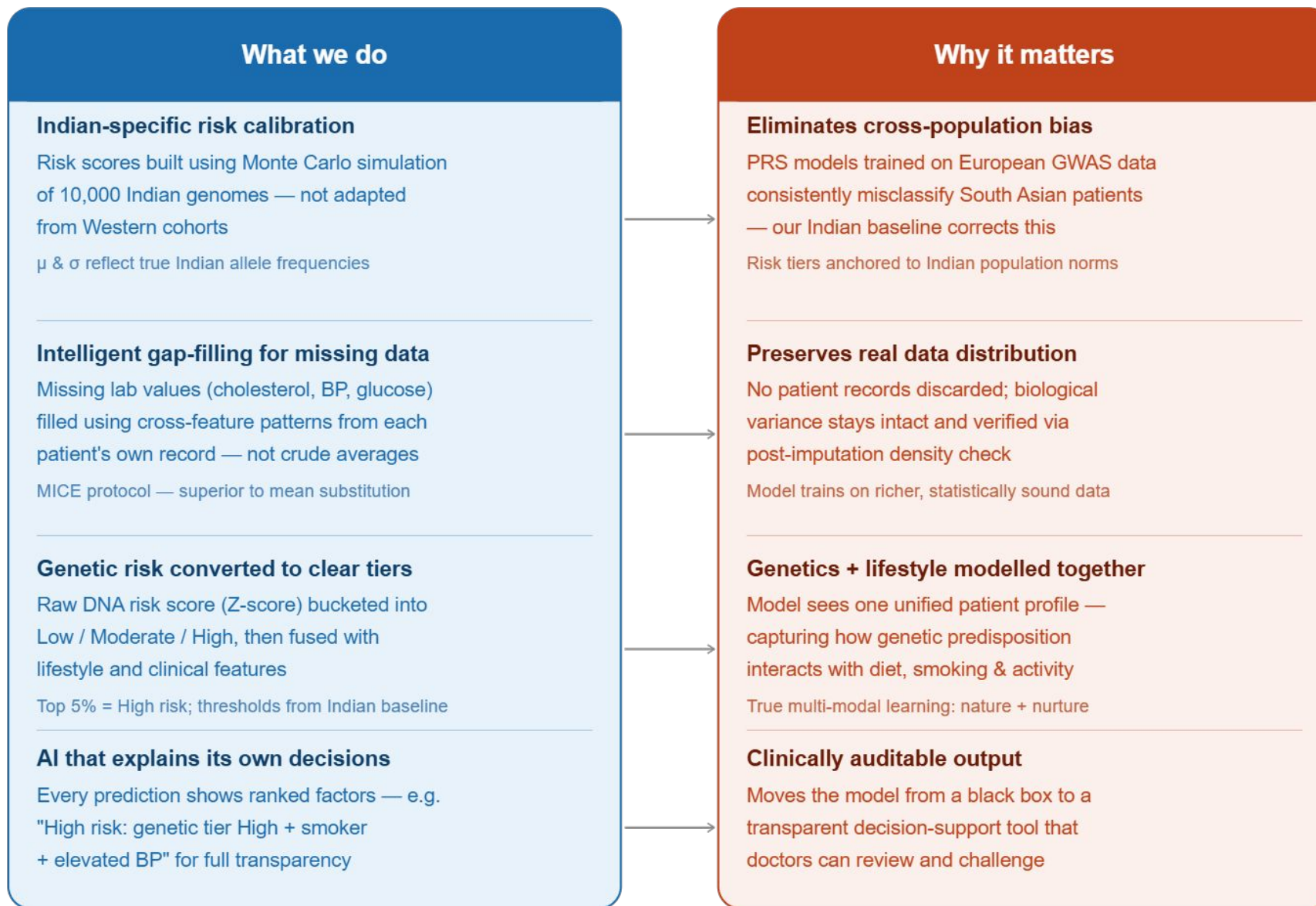
Proposed Methodology / Approach

Proposed Hybrid Methodology

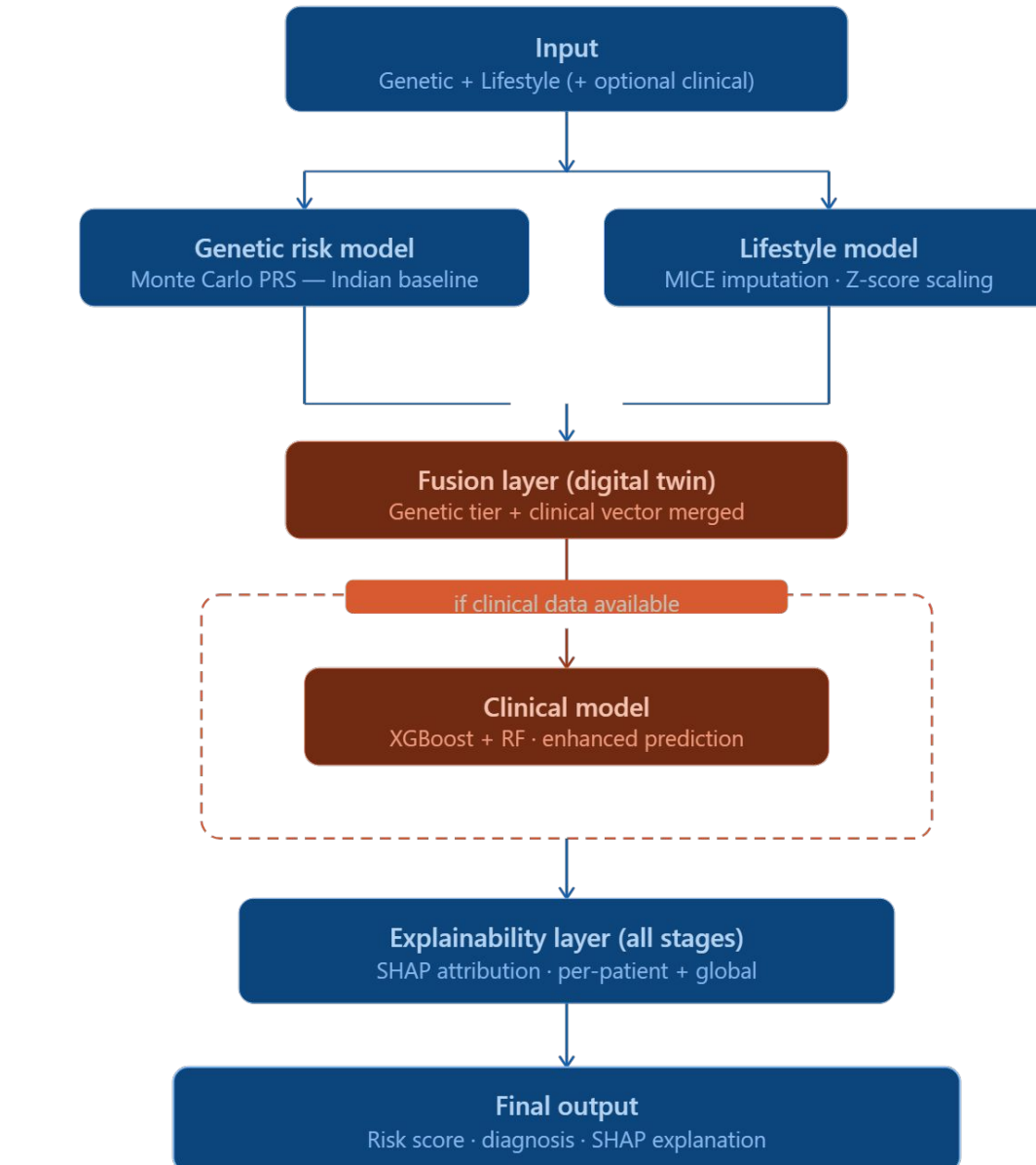
An end-to-end Nature × Nurture framework for CAD risk prediction in the Indian population

- Population-aware PRS calibration: PRS anchored to Indian allele frequencies via Monte Carlo simulation
- Dual-pipeline preprocessing: Genomic QC and clinical MICE imputation run in parallel, independently validated
- Feature-level fusion: Genetic risk tier concatenated with standardized clinical vector before model training
- Explainable ML prediction: XGBoost/RF outputs interpreted via per-feature SHAP attribution scores

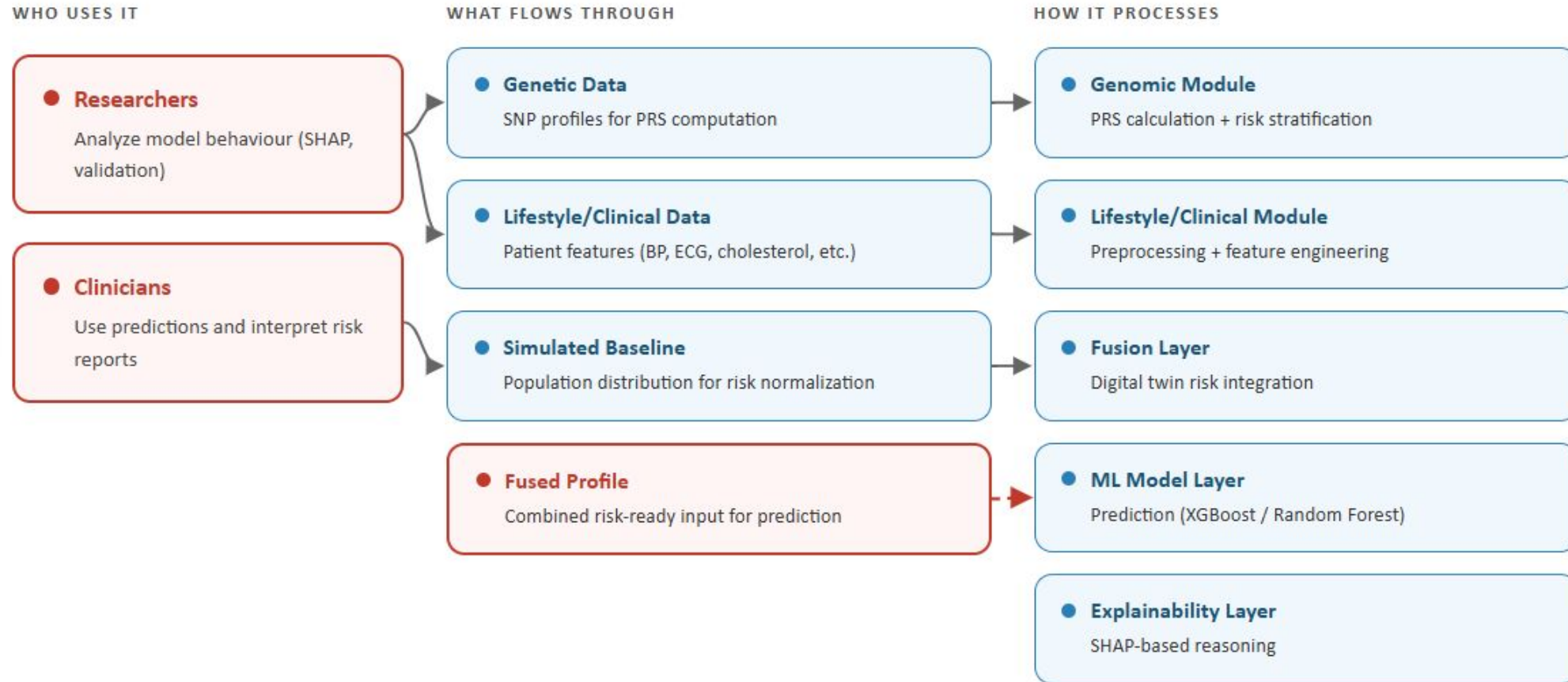
Proposed Methodology / Approach



System Architecture



Architecture Components



Data Exploration

Attribute	Genome India (GI)	PGS Catalog (PGS)	IEEE Cardiovascular Dataset (Integrated Clinical Datasets)	Lifestyle Dataset
Source	Genome India Project – Indian Biological Data Centre (IBDC)	Polygenic Score Catalog	IEEE DataPort Cardiovascular Dataset	Clinical lifestyle / heart disease dataset
Data Size	~129.9 million variants across 22 TSV files	205 SNP variants	~70,000 records (expanded dataset) + ~303 (Cleveland) + ~294 (Hungarian)	1,190 patient records
Data Type	Genomic variant data (hg38 reference build)	Polygenic score SNP dataset	Clinical + lifestyle + physiological cardiovascular data (multi-source integrated dataset)	Clinical + lifestyle data
Key Columns / Variables	Chromosome, Position, ID, Reference Allele, Alternate Allele, Alternate Allele Frequency	rsID, Chromosome Name, Chromosome Position, Effect Allele, Effect Weight, Allele Frequency Effect, Locus Name, Odds Ratio	Age, Sex, Chest Pain, BP, Cholesterol, Blood Sugar, ECG, Heart Rate, Angina, Oldpeak, Slope, CA, Thal, Target + lifestyle factors	Age, Sex, Chest Pain Type, Resting BP, Cholesterol, Fasting Blood Sugar, Resting ECG, Max Heart Rate, Exercise Angina, Oldpeak, ST Slope, Target

Data Exploration: Lifestyle Data

Dataset Overview:

- 70,000 instances, 12 features
- Mix of demographic, physiological, and lifestyle variables
- Binary target: presence/absence of cardiovascular disease

Key Features:

- **Demographic:** age, gender
- **Physical:** height, weight → BMI (derived)
- **Clinical:** blood pressure (ap_hi, ap_lo), cholesterol, glucose
- **Lifestyle:** smoking, alcohol, physical activity

	A	B	C	D	E	F	G	H	I	J	K	L
1	age	gender	height	weight	ap_hi	ap_lo	cholesterol	gluc	smoke	alco	active	target
2	50.39178	m	168	62	110	80	1	1	0	0	1	0
3	55.41918	f	156	85	140	90	3	1	0	0	1	1
4	51.66301	f	165	64	130	70	3	1	0	0	0	1
5	48.28219	m	169	82	150	100	1	1	0	0	1	1
6	47.87397	f	156	56	100	60	1	1	0	0	0	0
7	60.03836	f	151	67	120	80	2	2	0	0	0	0
8	60.58356	f	157	93	130	80	3	1	0	0	1	0
9	61.87397	m	178	95	130	90	3	3	0	0	1	1
10	48.40548	f	158	71	110	70	1	1	0	0	1	0

Data Exploration: Clinical Data

Dataset Overview

- 1190 instances | 11 features
- Integrated dataset from multiple sources (Cleveland, Hungarian, etc.)
- Target: Heart Disease (0 = No, 1 = Yes)

Key Feature Types

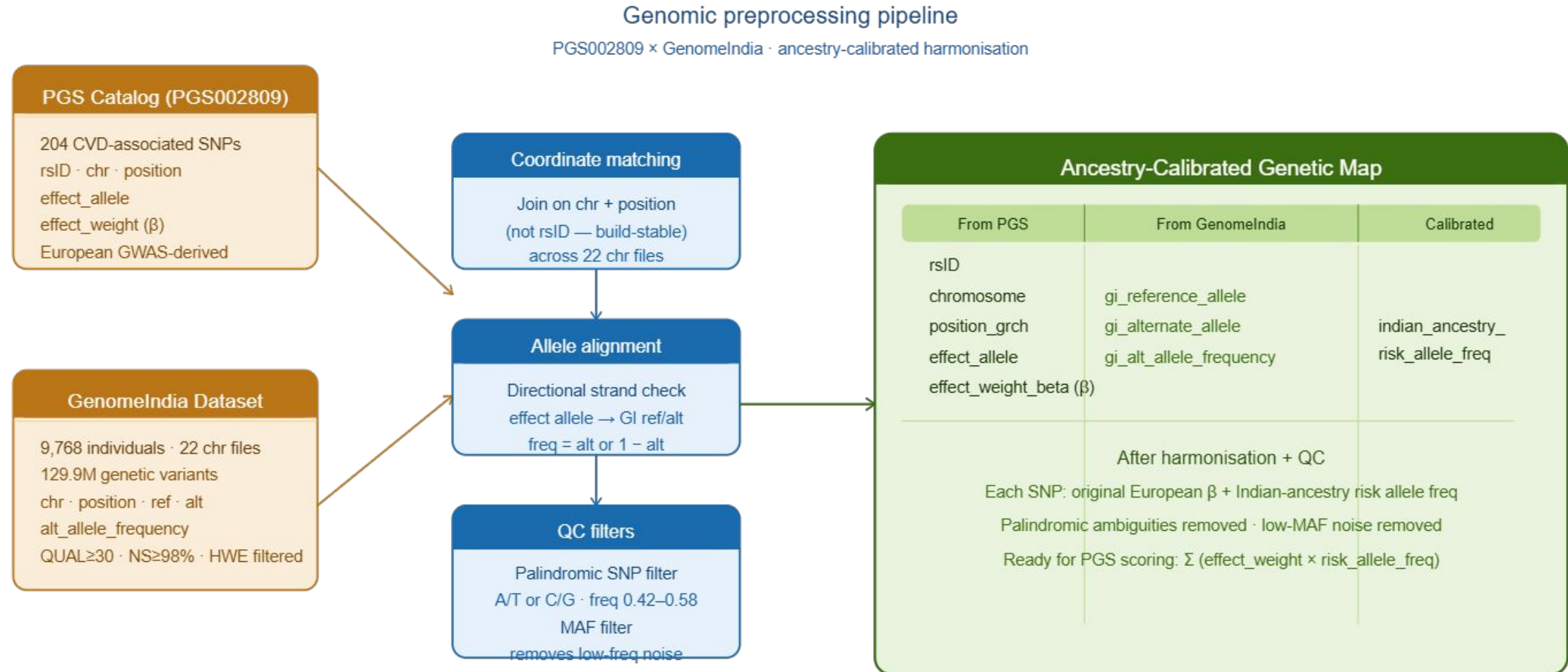
- **Numerical:** age, resting BP, cholesterol, max heart rate, oldpeak
- **Categorical:** chest pain type, ECG results, ST slope
- **Binary:** sex, fasting blood sugar, exercise-induced angina

	A	B	C	D	E	F	G	H	I	J	K	L
1	age	sex	chest pain type	resting bp s	cholesterol	fasting blood sugar	resting ecg	max heart rate	exercise angina	oldpeak	ST slope	target
2	40	1	2	140	289	0	0	172	0	0	1	0
3	49	0	3	160	180	0	0	156	0	1	2	1
4	37	1	2	130	283	0	1	98	0	0	1	0
5	48	0	4	138	214	0	0	108	1	1.5	2	1
6	54	1	3	150	195	0	0	122	0	0	1	0
7	39	1	3	120	339	0	0	170	0	0	1	0
8	45	0	2	130	237	0	0	170	0	0	1	0
9	54	1	2	110	208	0	0	142	0	0	1	0
10	37	1	4	140	207	0	0	130	1	1.5	2	1

Plan and progress on data preprocessing

Component	Data Source	Current Status
Genetic Data	PGS Catalog (204 SNPs) + Genomelndia Dataset (129.9M variants, ~9,768 individuals)	Completed
Lifestyle Data	IEEE DataPort Cardiovascular Disease Dataset (70,000 instances, 12 features) <i>(Lifestyle dataset with demographic, physiological, and behavioral risk factors)</i>	In Progress
Clinical Data	IEEE DataPort Comprehensive Heart Disease Dataset (1190 instances, 11 features) <i>(Curated by combining Cleveland, Hungarian, Switzerland, Long Beach VA, and Statlog datasets)</i>	In Progress

Data Preprocessing: Genetic Data Preprocessing



Data Preprocessing: Genetic Data Preprocessing

- Coordinate-Based SNP Matching**
 Join PGS (204 SNPs) with GenomeIndia (129.9M variants) using chromosome + position
- Allele Alignment (Strand Correction)**
 Adjust allele direction using reference/alternate mapping to prevent risk inversion
- Quality Control Filtering**
 Remove ambiguous (A/T, C/G) SNPs and low MAF variants
- Ancestry Calibration Output**
 Mapping with Indian allele frequencies using GenomeIndia for accurate PRS

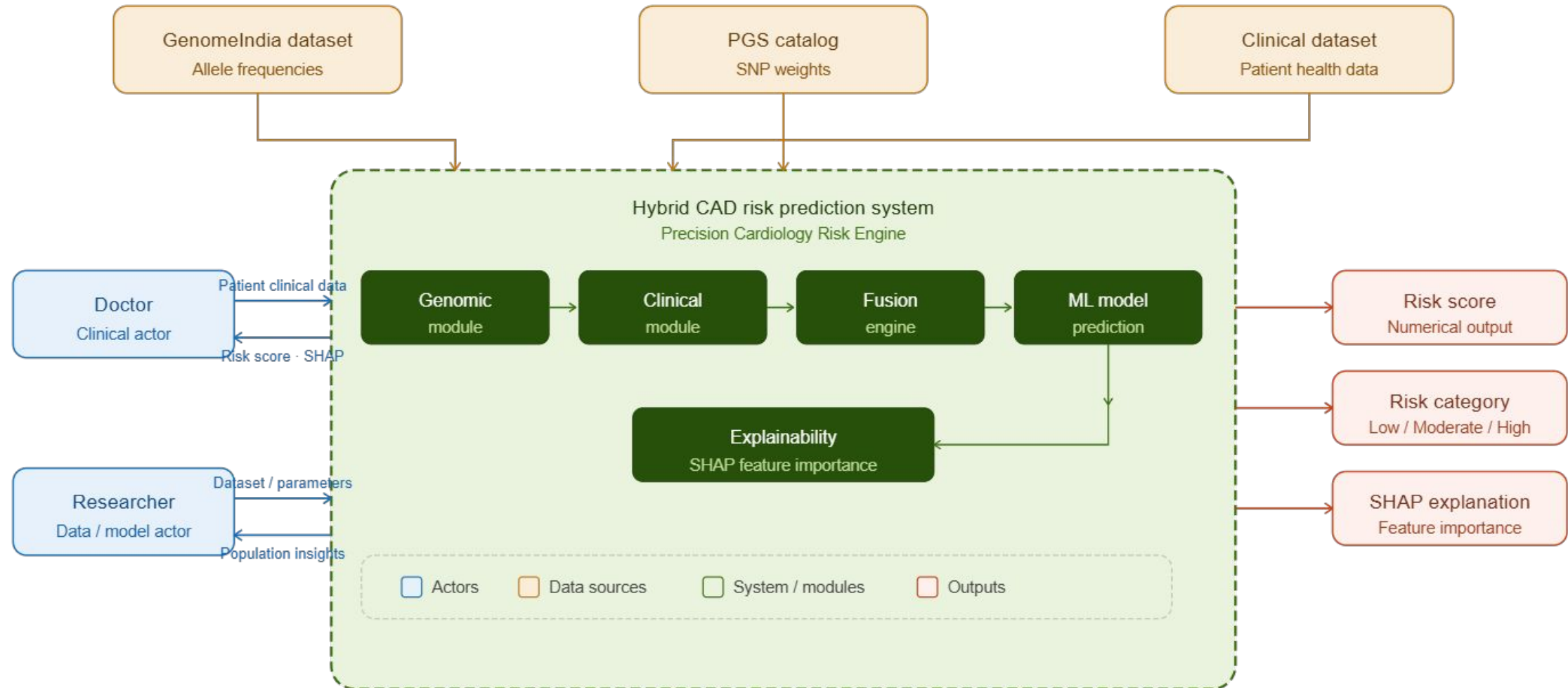
	A	B	C	D	E	F	G	H	I
1	rsID	chromosome	position_grch	effect_allele	effect_weight_beta	gi_reference_allele	gi_alternate_allele	gi_alt_allele_frequency	indian_ancestry_risk_allele_freq
2	rs36096196	1	2320766	T	0.04689999989	C	T	0.168765	0.168765
3	rs2493298	1	3409348	A	0.05140000034	C	A	0.0516324	0.0516324
4	rs61776719	1	37995647	A	0.04129999994	C	A	0.419107	0.419107
5	rs11206510	1	55030366	T	0.06765864847	T	C	0.0478457	0.9521543
6	rs11591147	1	55039974	G	0.2406000002	G	T	0.000409333	0.999590667
7	rs9970807	1	56499992	C	0.1142211441	C	T	0.0637089	0.9362911
8	rs61772618	1	56532813	C	0.07603468628	T	C	0.0522518	0.0522518
9	rs12740374	1	109274968	G	0.1238999999	G	T	0.246059	0.753941
10	rs11806316	1	115210861	G	0.0337000001	G	A	0.125077	0.874923

Technologies and Tool Exploration

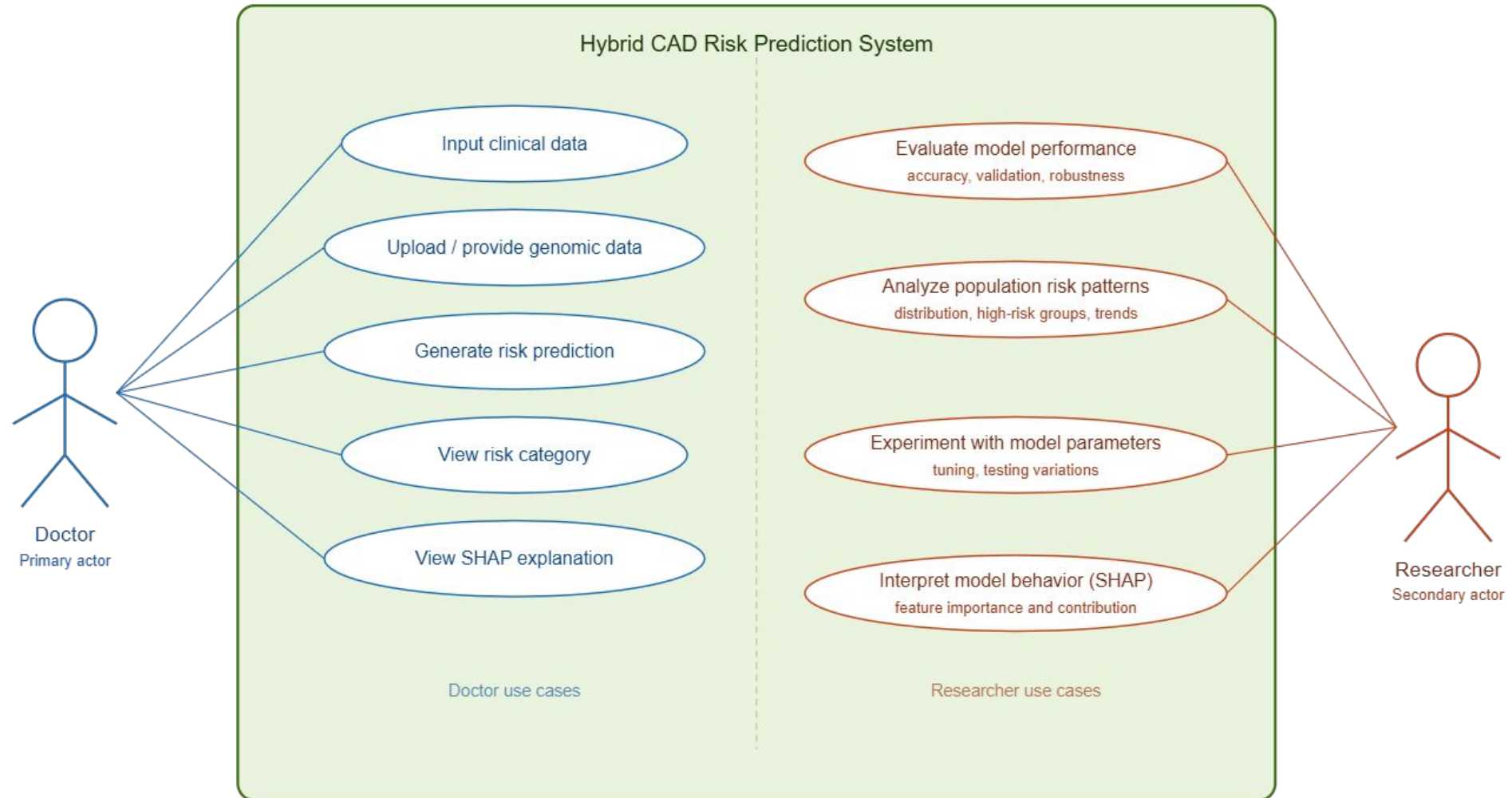
Tools and framework as applicable to your project

Pipeline Stage	Approach	Why This Was Chosen
Genomic Processing	Monte Carlo simulation — 10,000 virtual Indian genomes → Low / Mid / High tiers	Captures Indian population variability; avoids one-size-fits-all scoring
Missing Data Handling	MICE (IterativeImputer) — fills each gap using the patient's other values	Preserves real data relationships; no records discarded or distorted
Feature Scaling	Z-score Normalisation — Mean = 0, Std Dev = 1 across all variables	Stops large-range features dominating; e.g. Age (0–80) vs Oldpeak (0–6)
Model Selection	Random Forest + XGBoost — ensemble and boosting models run in parallel	Handles non-linear interactions; proven on cardiac ML; works on ~1,000 samples
Validation Framework	Stratified K-Fold CV — every record tested at least once across K splits	Prevents overfitting; maintains class balance; reflects real-world performance
Explainability	SHAP Values — per-patient and global explanations for every prediction	Explains each individual prediction; aligns with how clinicians reason

Design Description: External Interfaces



Design Description: Use Case Diagrams



References

- [1]** A. V. Khera, M. Chaffin, K. G. Aragam, M. E. Haas, C. Roselli, S. H. Choi, et al., “Genome-wide polygenic scores for common diseases identify individuals with risk equivalent to monogenic mutations,” *Nature Reviews Cardiology*, vol. 19, no. 4, pp. 225–238, 2022.
- [2]** A. G. Hindy, P. Natarajan, and S. Kathiresan, “Ancestry-specific polygenic risk scores and improvement of coronary artery disease prediction,” *Nature Communications*, vol. 14, Art. no. 1023, 2023.
- [3]** GenomeIndia Consortium, “The GenomeIndia Project: Cataloguing genetic diversity of the Indian population,” *Nature Genetics*, vol. 57, pp. 1–10, 2025.
- [4]** S. Corral-Acero, F. Margara, E. Marciniak, et al., “The ‘Digital Twin’ to enable the vision of precision cardiology,” *npj Digital Medicine*, vol. 3, Art. no. 1, pp. 1–10, 2022.

Thank You